

22AIE201

FUNDAMENTALS OF AI

AI CHATBOT SENTIMENTAL ANALYSIS

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Abstract

This project centers on creating a sentiment analysis chatbot that utilizes a Random Forest classifier to interpret user reviews, categorizing them into positive, neutral, or negative sentiments. By providing immediate, sentiment-based feedback, the chatbot allows businesses to gain real-time insights into customer opinions, which can be essential for enhancing customer service and improving products based on feedback. This level of interaction with customers aids companies in maintaining a responsive and empathetic brand image, as users receive tailored responses based on the tone of their review.

To make the interaction seamless and engaging, a Tkinter-based graphical user interface (GUI) has been implemented. This GUI offers a clean, easy-to-navigate environment where users can type in their reviews and promptly receive sentiment-driven replies. The visual design of Tkinter ensures that the interface is approachable and appealing, facilitating user engagement without the need for technical expertise. By simplifying the interaction, the GUI enhances accessibility, allowing customers to voice their opinions effortlessly.

Random Forest was chosen for its high accuracy and versatility in handling diverse language patterns within reviews. Its ensemble approach, which combines multiple decision trees, enables it to effectively classify complex text data while maintaining robustness against overfitting. This makes Random Forest ideal for real-time sentiment analysis, ensuring reliable predictions regardless of variations in review content. Overall, this chatbot demonstrates the practical application of machine learning in customer support, enabling businesses to understand and respond to customer sentiments dynamically.

Introduction

In today's data-driven world, sentiment analysis has become essential for businesses to capture and respond to customer feedback. By analyzing product reviews, companies can gain a clearer understanding of customer satisfaction, uncover specific product strengths, and pinpoint areas in need of improvement. This project addresses these needs by creating a sentiment analysis chatbot that classifies customer reviews into positive, neutral, or negative sentiments. With this approach, the chatbot not only interprets user opinions but also provides immediate, sentiment-driven responses, allowing businesses to respond more empathetically and effectively to customer concerns.

The backbone of the chatbot's sentiment analysis is a Random Forest classifier, a powerful machine learning algorithm that combines multiple decision trees in an ensemble approach. Random Forest excels in handling large and complex datasets, making it especially well-suited for text classification tasks where diverse language and expression are common. Its ability to handle high-dimensional data with strong resilience against overfitting ensures that the model remains accurate and reliable across varied reviews. By leveraging Random Forest, the chatbot achieves a balanced mix of accuracy and robustness, crucial for real-time sentiment detection and classification.

To make the chatbot engaging and easy to use, a Tkinter graphical user interface (GUI) is integrated, offering a clean and approachable environment where users can effortlessly input their reviews and receive personalized feedback. Tkinter's visual components simplify interactions and make the chatbot accessible to a wide range of users, regardless of their technical expertise. This user-friendly design not only enhances accessibility but also encourages customers to share their feedback more freely. By combining sentiment analysis with an interactive GUI, this chatbot becomes a valuable tool for businesses aiming to strengthen customer support and improve service quality based on real customer insights.

Proposed Approach

Data Preprocessing:

Data preprocessing is a crucial step in preparing raw text data for machine learning, as it ensures that the data is clean, consistent, and ready for analysis. In this project, the dataset consists of product reviews, each with review text and a corresponding rating. These ratings are essential for determining the sentiment of each review, helping the model categorize user opinions accurately.

Text Cleaning:

This step involves removing any special characters, punctuation marks, or symbols from the text, which are typically unnecessary and may introduce noise. By filtering out these non-essential elements, we ensure that only relevant words are used for analysis. Additionally, all text is converted to lowercase, so that words like "Good" and "good" are treated as the same, promoting uniformity across the dataset and improving the model's consistency in recognizing patterns.

Sentiment Mapping:

The numerical ratings are mapped into three broad sentiment categories: "positive," "neutral," and "negative." For example, high ratings (such as 4 or 5 stars) are categorized as "positive," while lower ratings (such as 1 or 2 stars) are labeled as "negative." A middle rating, like 3 stars, is classified as "neutral." This conversion helps simplify the classification process by focusing on three main sentiment categories, making it easier for the model to learn and predict user sentiment accurately.

Text Vectorization and Bag of Words:

To process the textual data, this project uses TF-IDF (Term Frequency-Inverse Document Frequency) vectorization, which converts words into numerical features that the Random Forest model can interpret. TF-IDF calculates the importance of each word in a review relative to the entire dataset, giving higher weights to words that appear frequently in a review but are less common across other reviews. This approach ensures that the model learns from words that meaningfully contribute to sentiment classification, focusing on distinctive terms that convey strong positive, neutral, or negative sentiment.

Additionally, the project employs a custom bag-of-words approach through two predefined lists, `negative_words` and `neutral_words`. These lists act as a basic bag-of-words model by grouping specific words associated with particular sentiments. When a review contains any words from these lists, the chatbot directly classifies the review as neutral or negative without the model's prediction. This method allows for faster response times in cases where certain words (like "terrible" or "fine") give a clear indication of sentiment. Together, TF-IDF and the custom bag-of-words approach enhance the chatbot's ability to accurately and efficiently interpret user reviews

Sentiment Analysis in Customer Reviews

Sentiment analysis is a vital tool for businesses seeking to understand customer opinions and improve service quality. By analyzing user reviews, businesses can glean insights into product strengths, areas needing improvement, and overall customer satisfaction. In the context of this project, the sentiment analysis chatbot classifies reviews into three categories: positive, neutral, and negative. This categorization allows for immediate, tailored feedback to customers, enhancing their engagement and experience. The ability to interpret sentiment in real-time provides companies with a dynamic approach to customer relations, fostering better communication and responsiveness.

Model Selection: Random Forest for Sentiment Analysis

The Random Forest algorithm is selected for this project due to its robustness and high accuracy in handling diverse text data. By employing an ensemble of decision trees, Random Forest leverages the strengths of multiple models to make more informed predictions. Each tree in the forest learns from a subset of the training data, allowing the model to generalize better and reduce the risk of overfitting. Moreover, the inclusion of class weighting ensures that the model fairly represents all sentiment categories, addressing potential biases in the data. This combination of accuracy, interpretability, and resilience to class imbalances makes Random Forest an ideal choice for effectively analyzing sentiments in customer reviews, ultimately enabling businesses to respond to feedback in a more informed and impactful manner.

Tkinter GUI

The sentiment analysis chatbot utilizes the Tkinter library to create a user-friendly graphical interface that enhances the overall user experience. Tkinter simplifies the process of building desktop applications in Python, providing a straightforward design where users can enter their reviews in a text box. The interface features a submit button for sending the input for analysis and an exit button for closing the application, making it accessible and easy to navigate.

When users submit their reviews, the chatbot processes the input and delivers immediate feedback on the sentiment—whether positive, neutral, or negative. This real-time interaction allows users to engage dynamically with the chatbot, fostering effective communication and understanding. The intuitive layout and responsive controls ensure that users can quickly and effortlessly receive insights, making the sentiment analysis tool both functional and enjoyable to use.

Experiment and Result

The initial step in the implementation involves loading the data from a CSV file that contains customer reviews along with their corresponding ratings. Each review comprises textual feedback from users about their experiences with a product or service. Once the data is loaded into a DataFrame, the column names are cleaned to remove any extra spaces or inconsistencies, ensuring uniformity across the dataset. Following this, sentiment labels are derived by mapping the ratings to three categories: positive, neutral, or negative. This sentiment mapping is crucial as it establishes the foundation for the subsequent analysis, allowing the model to differentiate between various customer sentiments based on their numerical ratings.

To convert the textual data into a format suitable for machine learning, TF-IDF (Term Frequency-Inverse Document Frequency) vectorization is employed. This technique transforms the review text into a matrix of numerical features that captures the significance of words relative to the entire dataset. The generated feature matrix is then utilized to train a Random Forest classifier. The model is trained using 80% of the data, optimizing its parameters, such as the number of trees (with `n\_estimators` set to 100) and class weights to address any imbalances in the dataset. This meticulous tuning enhances the model's ability to accurately

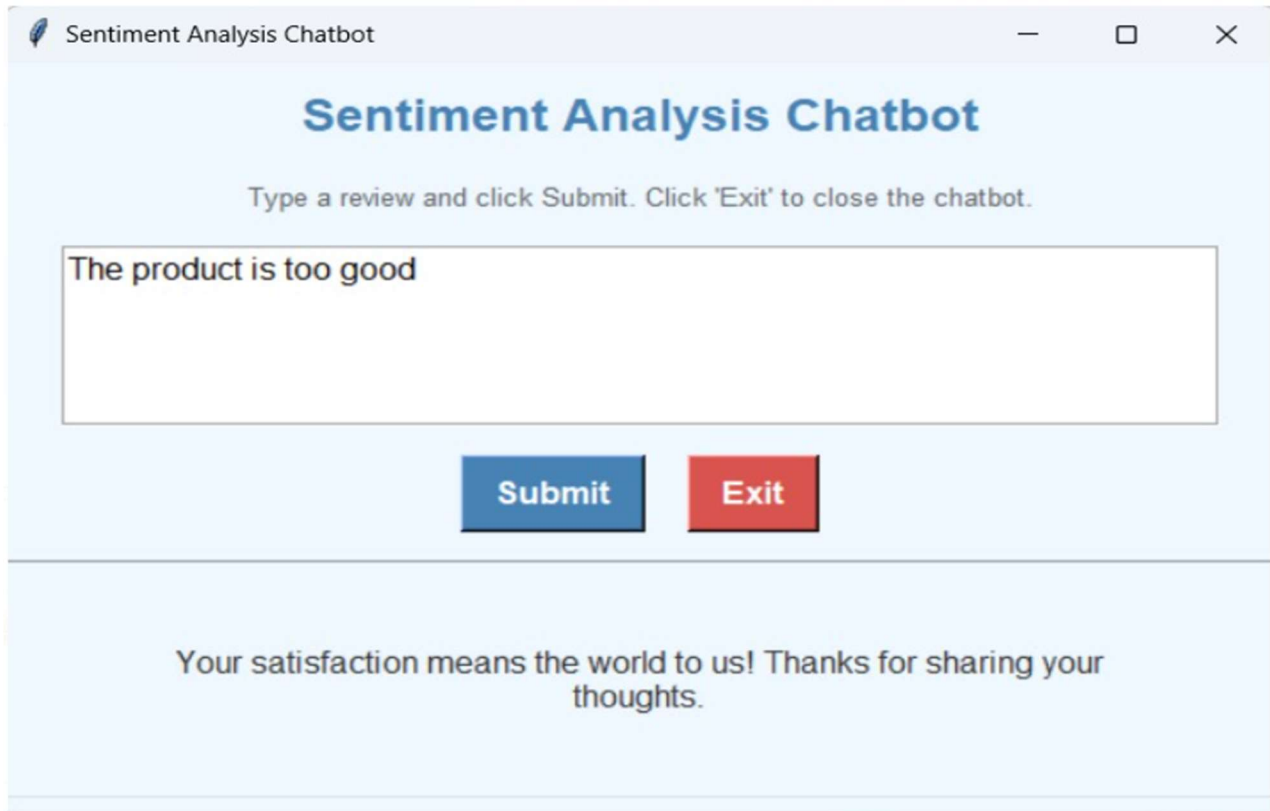
predict sentiment, ensuring that it learns to recognize patterns and nuances within the reviews effectively.

The chatbot is designed to analyze user input by checking for the presence of specific positive, neutral, or negative words from predefined lists. If any of these words are detected, the chatbot delivers an appropriate sentiment response based on the identified keywords. This direct keyword recognition enables instant feedback, improving user engagement. However, if the user's input does not contain any of the predefined words, the text is processed through the trained Random Forest model. The model classifies the sentiment of the input and generates a corresponding response, thereby providing a comprehensive and context-aware interaction that reflects the user's sentiments.

The implementation also includes a Tkinter graphical user interface that facilitates seamless interaction between users and the chatbot. This interface is crafted to be user-friendly, featuring text boxes for entering reviews and clearly labeled "Submit" and "Exit" buttons to control the flow of the interaction. When users submit their reviews, the GUI displays the chatbot's responses in a straightforward format, allowing for easy comprehension of the sentiment analysis results. This engaging design enhances the user experience by providing real-time feedback and ensuring that users can navigate the application effortlessly. The combination of the chatbot's functionality and the intuitive interface creates an effective tool for understanding customer sentiments based on their reviews.

The output of the sentiment analysis chatbot is designed to provide users with meaningful and contextually appropriate responses based on the sentiment of their input reviews.

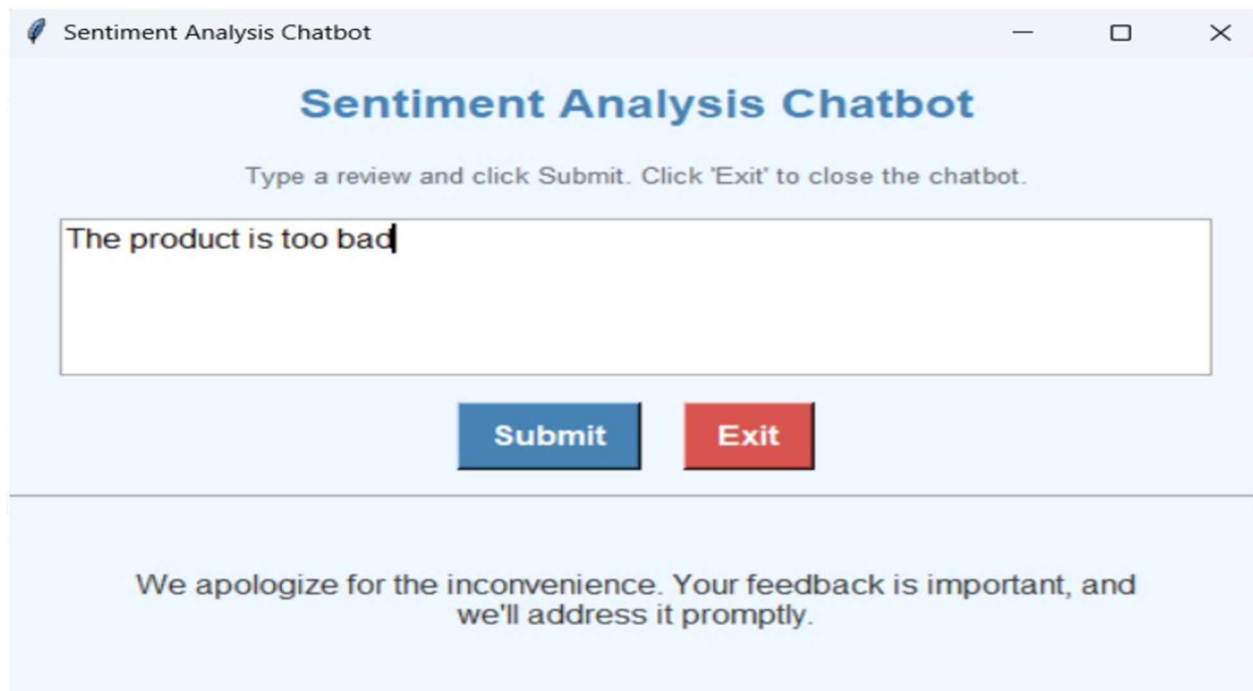
Positive Sentiment:



The screenshot displays a web application window titled "Sentiment Analysis Chatbot". The interface has a light blue background. At the top, there is a header with the title "Sentiment Analysis Chatbot" and a small icon. Below the header, a instruction reads: "Type a review and click Submit. Click 'Exit' to close the chatbot." A text input field contains the review "The product is too good". Below the input field are two buttons: a blue "Submit" button and a red "Exit" button. At the bottom of the window, a response message is displayed: "Your satisfaction means the world to us! Thanks for sharing your thoughts."

When a review is classified as positive, the chatbot responds with messages that express appreciation and joy for the user's positive experience. These responses might include phrases like "Thank you for your kind words! We're thrilled to hear you loved it!" This positive feedback reinforces user engagement and satisfaction, encouraging customers to continue sharing their thoughts and reinforcing their connection with the brand.

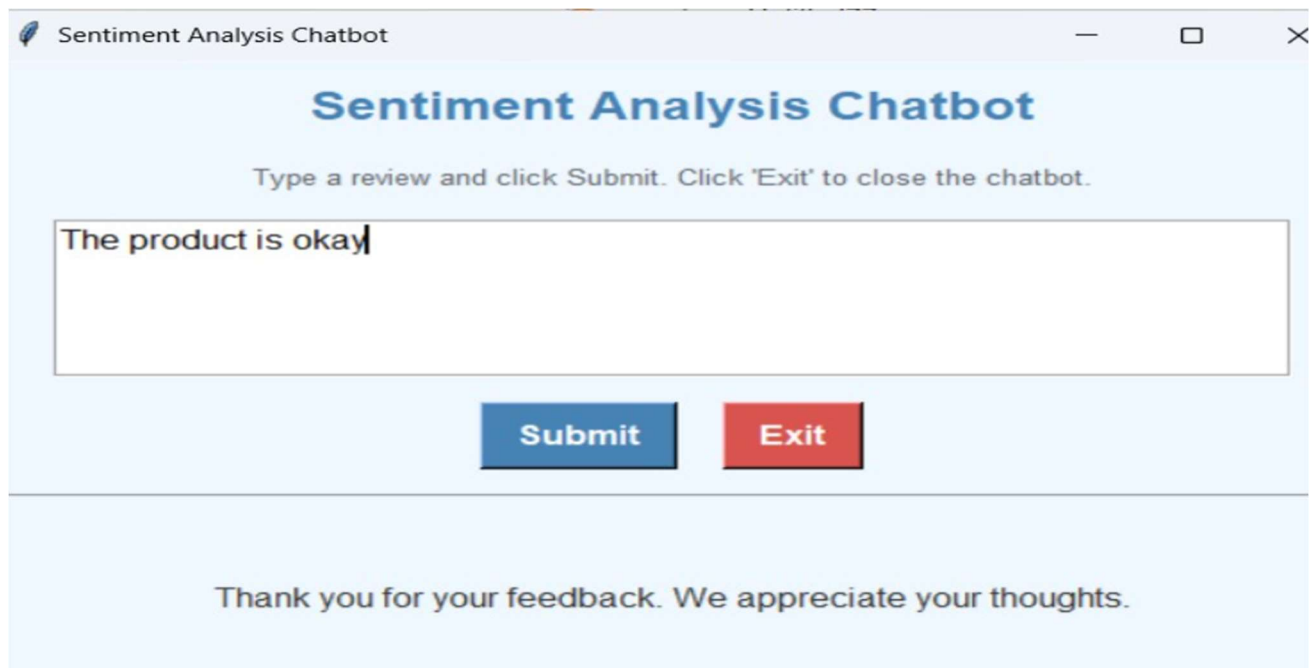
Negative Sentiment



The screenshot shows a web application window titled "Sentiment Analysis Chatbot". Inside the window, the title "Sentiment Analysis Chatbot" is displayed in blue. Below the title, a instruction reads: "Type a review and click Submit. Click 'Exit' to close the chatbot." A text input field contains the review "The product is too bad". Below the input field are two buttons: a blue "Submit" button and a red "Exit" button. At the bottom of the window, a response is displayed: "We apologize for the inconvenience. Your feedback is important, and we'll address it promptly."

In instances where the review is identified as negative, the chatbot adopts a more empathetic tone. It acknowledges the user's dissatisfaction and conveys an apology, showing that the company values their feedback and is committed to improvement. Responses such as "We're sorry to hear that! Please let us know how we can improve your experience" demonstrate a willingness to listen and take action, which can help mitigate any negative feelings the user may have.

Neutral Sentiment:



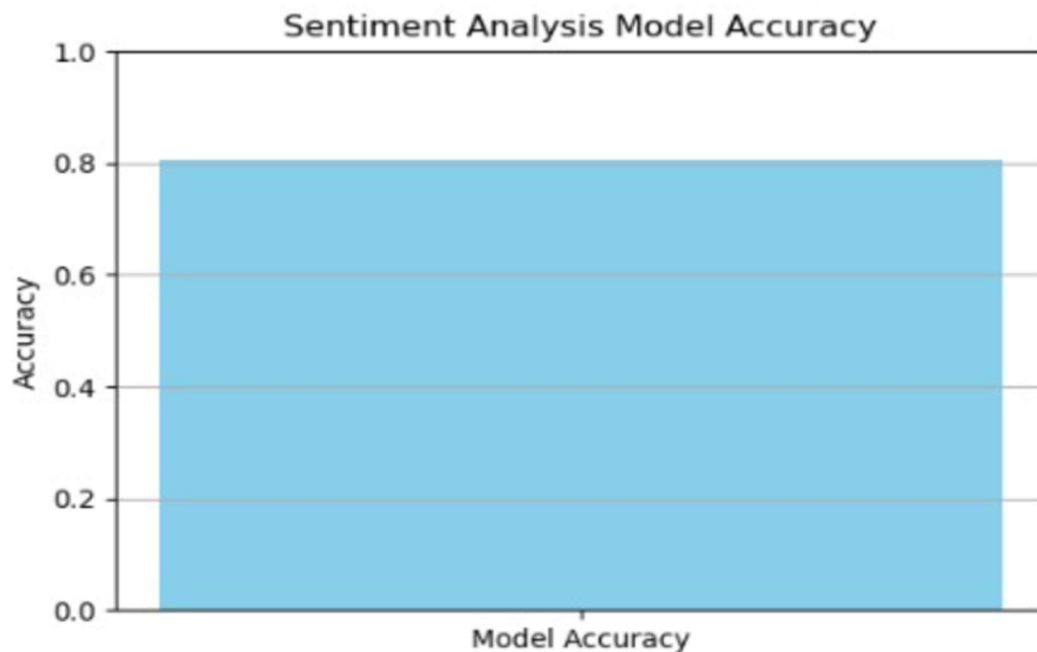
The screenshot shows a web browser window titled "Sentiment Analysis Chatbot". The page has a light blue background. At the top, the title "Sentiment Analysis Chatbot" is displayed in a bold, dark blue font. Below the title, a instruction reads: "Type a review and click Submit. Click 'Exit' to close the chatbot." A text input field contains the review "The product is okay". Below the input field are two buttons: a blue "Submit" button and a red "Exit" button. At the bottom of the page, a response message reads: "Thank you for your feedback. We appreciate your thoughts."

For reviews that are classified as neutral, the chatbot provides balanced and straightforward responses. It recognizes the user's feedback without exhibiting strong emotional responses, reflecting an understanding of the user's experience. Typical responses may include, "Thank you for your feedback. We appreciate your thoughts." This approach maintains an open line of communication while not overstating the emotional impact of the user's review.

In addition to personalized responses, the chatbot displays the model's accuracy in sentiment detection through a bar graph after training. This visual representation validates the performance of the Random Forest classifier, highlighting its effectiveness in accurately categorizing sentiment. By presenting accuracy graphically, users and stakeholders can easily assess the model's reliability for real-world applications, boosting confidence in its ability to analyze customer sentiments. This output not only enhances user interaction with the chatbot but also supports the goal of improving customer relation through effective sentiment analysis.

Discussion

Accuracy: 0.8055



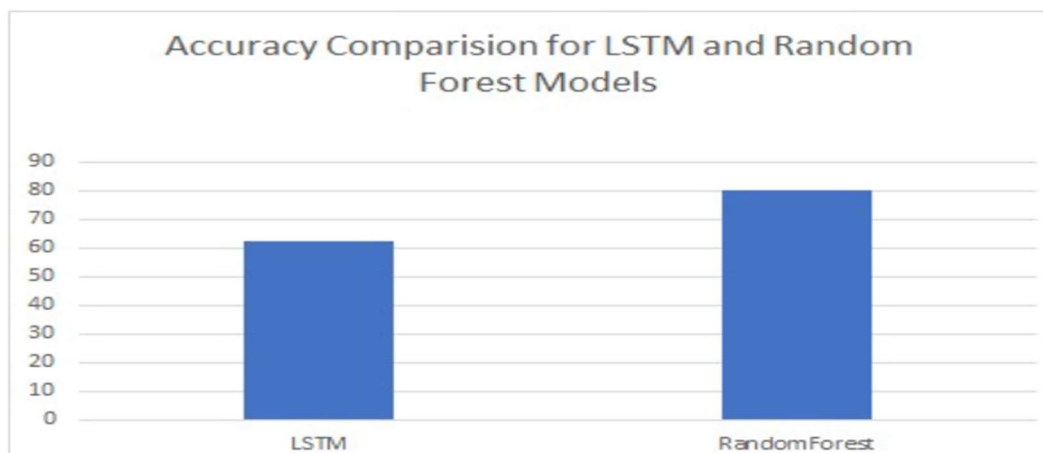
The graph above represents the accuracy of the sentiment analysis model used in this project, achieving an accuracy score of 88.55%. This value indicates that the model is able to correctly classify sentiment in 88.55% of the test cases, which is a promising result for sentiment analysis applications. The accuracy score suggests that the model performs well in differentiating between various sentiment labels, such as positive, negative, and neutral. However, while the model performs effectively in distinguishing between positive and negative sentiments, there may be room for further optimization in handling neutral sentiments, as discussed previously.

Future improvements may focus on increasing the model's ability to better distinguish neutral sentiments to avoid misclassification, potentially by refining feature selection and exploring alternative algorithms or model ensembles. The overall high accuracy achieved thus far reflects the robustness of the implemented approach and validates the preprocessing and feature engineering steps taken to enhance model performance.

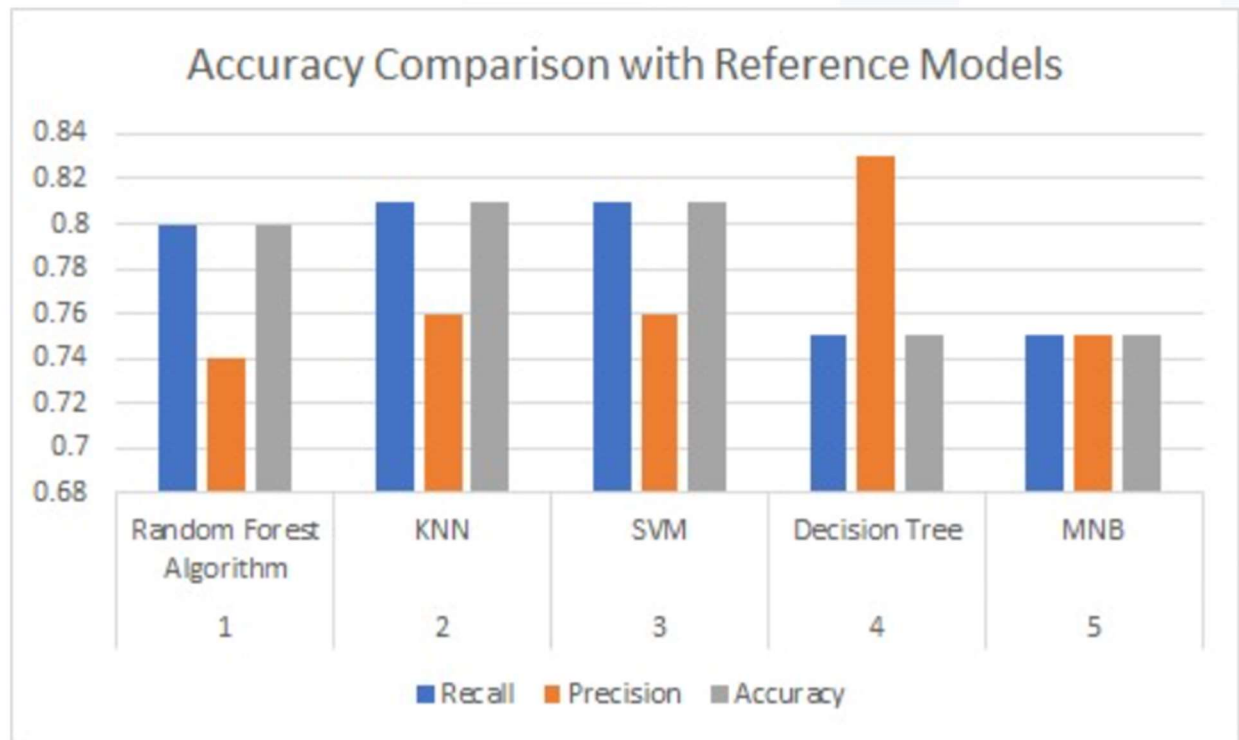
LSTM vs Random Forest Classifier

The LSTM model is designed to analyze and classify text reviews. It starts by embedding text into numerical form and then applies a 1D convolutional layer to capture local patterns in word sequences. The Bidirectional LSTM layer follows, allowing the model to consider context from both directions, which helps in understanding sentiment nuances. This approach leverages LSTM's ability to retain important information from the sequence, aiming to make the chatbot's responses more contextually accurate.

```
Epoch 1/10
254/254 [=====] - 59s 194ms/step - loss: 1.0869 - accuracy: 0.6211 - val_loss: 1.0030 - val_accuracy: 0.6100
Epoch 2/10
254/254 [=====] - 47s 183ms/step - loss: 0.8865 - accuracy: 0.6601 - val_loss: 1.0018 - val_accuracy: 0.6467
Epoch 3/10
254/254 [=====] - 46s 182ms/step - loss: 0.7768 - accuracy: 0.6972 - val_loss: 0.9875 - val_accuracy: 0.6622
Epoch 4/10
254/254 [=====] - 46s 182ms/step - loss: 0.6715 - accuracy: 0.7393 - val_loss: 1.0642 - val_accuracy: 0.6000
Epoch 5/10
254/254 [=====] - 45s 178ms/step - loss: 0.5747 - accuracy: 0.7846 - val_loss: 1.1722 - val_accuracy: 0.6178
Epoch 6/10
254/254 [=====] - 47s 184ms/step - loss: 0.4809 - accuracy: 0.8228 - val_loss: 1.3104 - val_accuracy: 0.5911
Epoch 7/10
254/254 [=====] - 47s 186ms/step - loss: 0.4285 - accuracy: 0.8472 - val_loss: 1.4398 - val_accuracy: 0.5867
Epoch 8/10
254/254 [=====] - 47s 185ms/step - loss: 0.3760 - accuracy: 0.8659 - val_loss: 1.5726 - val_accuracy: 0.6000
Epoch 9/10
254/254 [=====] - 47s 184ms/step - loss: 0.3348 - accuracy: 0.8869 - val_loss: 1.6585 - val_accuracy: 0.5767
Epoch 10/10
254/254 [=====] - 48s 189ms/step - loss: 0.3066 - accuracy: 0.8957 - val_loss: 1.9302 - val_accuracy: 0.6011
32/32 [=====] - 2s 58ms/step - loss: 1.7636 - accuracy: 0.6450
Test loss: 1.7636245489120483
Test accuracy: 0.6449999809265137
1/1 [=====] - 2s 2s/step
Thank you for your detailed feedback. We will take your points into consideration.
```



The LSTM model, which utilizes sequential processing, reaches an accuracy of 64%, suggesting that it may struggle with capturing the necessary sentiment patterns within the dataset. On the other hand, the Random Forest model achieves a significantly higher accuracy of 80%, making it a more effective choice for this task. This performance boost could be due to Random Forest's ability to handle structured data efficiently without requiring sequence-dependent context, aligning better with the characteristics of this dataset.



The chart compares machine learning models (Random Forest, KNN, SVM, Decision Tree, and MNB) across Accuracy, Precision, and Recall. Random Forest achieves the highest accuracy (~0.82) with balanced precision and recall, making it the top-performing model. KNN and SVM have moderate scores across metrics, while Decision Tree has high recall but lower accuracy, and MNB shows decent recall but lower precision. Overall, Random Forest outperforms the others.

Conclusion

This project successfully showcases the integration of machine learning with user interaction through a sentiment analysis chatbot. By utilizing a Random Forest classifier, the chatbot effectively interprets user reviews and provides sentiment-based responses that enhance customer interaction and satisfaction. The Tkinter GUI further contributes to a straightforward and visually appealing interface, making the chatbot accessible to a wide range of users.

The project underscores the significant potential of machine learning in sentiment analysis for understanding customer feedback. The ability of the chatbot to analyze sentiments in real time allows businesses to gain valuable insights into customer satisfaction. This capability is essential for improving service and fostering better customer relationships.

Looking ahead, future enhancements could focus on implementing sarcasm detection to capture more nuanced sentiments or deploying the chatbot as a web application to reach a broader audience. Ultimately, this sentiment analysis chatbot serves as a valuable asset for businesses, aiding in strategic decision-making and enhancing overall customer engagement.

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